

HYDROGENATED NITRILE BUTADIENE RUBBER (HNBR): HYDROGENATION ROUTES, CATALYST SYSTEMS, AND POST-TREATMENT TRENDS ACROSS SELECTED PATENTS

1. ABSTRACT

This report synthesizes nine primary patents focused on hydrogenated nitrile butadiene rubber (HNBR) and closely related hydrogenation/manufacturing routes. The selected landscape spans direct hydrogenation of NBR latex in water, solution-phase hydrogenation of NBR in organic solvent, simultaneous hydrogenation/metathesis routes that reduce molecular weight while saturating residual double bonds, and downstream purification/removal of noble-metal residues from HNBR solutions. Across the set, the target material is consistently HNBR or an HNBR precursor intended to yield HNBR, with the key target attribute being degree of hydrogenation or residual double-bond content. The patents collectively show that high hydrogenation levels are achievable by tuning catalyst identity, catalyst loading, temperature, hydrogen pressure, and reaction time, while avoiding gel formation and preserving processability.

A major process split is evident. One cluster uses aqueous latex hydrogenation without organic solvent, exemplified by ruthenium/osmium-type catalysts in water-containing polymer dispersions and by direct hydrogenation of NBR latex with Hoveyda-Grubbs-type catalysts. Another cluster uses monochlorobenzene or other organic solvents for solution hydrogenation, often with Ru, Rh, or Pd catalysts and sometimes with phosphine co-catalysts. A third cluster combines hydrogenation with metathesis to lower molecular weight and narrow molecular-weight distribution, using Ru or Os complexes designed to promote simultaneous chain scission and saturation. The final cluster addresses post-hydrogenation cleanup, especially removal of ruthenium residues by functionalized ion-exchange resins, reflecting the importance of catalyst residue control for aging performance and product quality.

Formulation and process trends include use of low ppm noble-metal catalyst levels, operation at elevated temperatures (roughly 100-155°C in the examples), hydrogen pressures from tens of bar up to about 1500 psi in latex systems, and careful control of residual unsaturation to below about 5% in several solution examples. Water is central only in the latex-based patent; in the others it is absent from the core reaction medium and appears, when present, only in analytical or downstream handling contexts. Overall, the patent set shows a mature HNBR technology space

balancing saturation efficiency, molecular-weight control, and catalyst-residue management.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US9657113B2
- Patent Title: Hydrogenation of a diene-based polymer latex
- Assignee: University Of Waterloo
- Jurisdiction: US
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses direct hydrogenation of NBR latex in an aqueous medium using solid Hoveyda-Grubbs second-generation catalyst (HG2) or related Grubbs catalyst systems. The process is run without organic solvent, with hydrogen introduced under pressure while the latex is heated, and the hydrogenation degree is monitored over time. - NBR solid content: 19.5 wt % in water — Table 1 - Butadiene content: 66% — Table 1 - Acrylonitrile content: 34% — Table 1 - Mean particle diameter: about 72 nm — Table 1 - NBR latex volume: 25 ml — Example 1 - Water added: 75 ml — Example 1 - HG2 catalyst: 0.00487 g — Example 1 - Nitrogen degassing: about 200 psi (1.38 MPa) — Example 1 - Nitrogen degassing time: 20 min — Example 1 - Room temperature: room temperature — Example 1 - Reaction temperature: 100° C. — Example 1 - Hydrogen pressure: 1000 psi (6.89 MPa) — Example 1 - Reaction time: 4 hours — Example 1 - Hydrogenation degree: 98% — Example 1 - HG2 catalyst: 0.00244 g — Example 2 - Reaction time: 19 hours — Example 2 - Hydrogenation degree: 90% — Example 2 - Reaction temperature: 120° C. — Example 3 - Reaction time: 1.6 hours — Example 3 - Hydrogenation degree: 98% — Example 3 - HG2 catalyst: 0.00975 g — Example 4 - Reaction time: 0.5 hours — Example 4 - Hydrogenation degree: 98% — Example 4 - HG2 catalyst: 0.00125 g — Example 5 - Reaction time: 10 hours — Example 5 - Hydrogenation degree: 74% — Example 5 - Hydrogen pressure: 500 psi (3.45 MPa) — Example 6 - Reaction time: 10 hours — Example 6 - Hydrogenation degree: 99% — Example 6 - Hydrogen pressure: 1500 psi (10.34 MPa) — Example 7 - Reaction time: 9 hours — Example 7 - Hydrogenation degree: 99% — Example 7 - NBR latex volume: 100 ml — Example 8 - HG2 catalyst: 0.01253 g — Example 8 - No extra water added — Example 8 - Reaction time: 3 hours — Example 8 - Hydrogenation degree: 99% — Example 8 - Hydrogenation degree: 15% — Comparative Example 1a - Hydrogenation degree: 22% — Comparative Example 1a - Hydrogenation degree: 249,000 — Example 1 - PDI: 3.1 — Example 1 - Medium & Water Role: The core hydrogenation medium is an aqueous NBR latex suspension; water is part of the

reaction medium, and in some examples additional water is added for dilution. No separate downstream water workup is emphasized in the extracted evidence. - degree of hydrogenation: 98%

Medium & Water Role - The core hydrogenation medium is an aqueous NBR latex suspension; water is part of the reaction medium, and in some examples additional water is added for dilution. No separate downstream water workup is emphasized in the extracted evidence. - Core reaction medium: Aqueous latex / aqueous suspension hydrogenation

degree of hydrogenation - 98%

Relevant Experimental Evidence - Example 1: Direct latex hydrogenation reached 98% hydrogenation in 4 hours with no visible gel and produced soluble HNBR. - Example 2: Lower HG2 loading still gave 90% hydrogenation after 19 hours without gel. - Example 3: Raising temperature to 120° C. shortened the time to 98% hydrogenation to 1.6 hours. - Example 6: Lower hydrogen pressure still gave 99% hydrogenation after 10 hours without gel. - Example 8: Higher latex concentration without extra water still gave 99% hydrogenation in 3 hours.

Technical Relevance This is a core HNBR synthesis patent because it directly converts NBR latex to HNBR and quantifies hydrogenation extent under aqueous, solvent-free conditions. It is especially relevant for understanding how catalyst loading, temperature, and hydrogen pressure control saturation while suppressing gel formation.

Patent 2

- Patent Number: US20210340285A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: Arlanxeo Deutschland Gmbh
- Jurisdiction: US
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses hydrogenation of at least partially unsaturated nitrile rubber in solution using ruthenium, palladium, or rhodium compounds, with the stated goal of producing HNBR having improved ageing properties. The evidence emphasizes catalyst residue management and solution-phase hydrogenation conditions rather than a specific worked example hydrogenation recipe. - Ruthenium amount: 10 ppm to 200 ppm — Claim 1 - Ruthenium amount: 10 ppm to 150 ppm — Claim 1 - Ruthenium amount: 10 ppm to 120 ppm — Claim 1 - Ruthenium amount: 10 ppm to 79 ppm — Claim 1 - Ruthenium amount: 42 ppm to 79 ppm — Claim 1 - Palladium amount: 20 ppm to <67 ppm — Claim 1 - Palladium amount: 44 ppm to <67 ppm — Claim 1 - Rhodium amount: 50 ppm to <270 ppm — Claim 1 - Rhodium amount: 79 ppm to <270 ppm — Claim 1 - Rhodium amount: 50 ppm to 250 ppm —

Claim 1 - Rhodium amount: 50 ppm to 170 ppm — Claim 1 - Rhodium amount: 79 ppm to 170 ppm — Claim 1 - Hydrogenation temperature: 60 to 200° C. — Claim 1 - Hydrogenation pressure: 700 000 pascals to 15 000 000 pascals — Claim 1 - Hydrogenation time: 1 to 24 hours — Claim 1 - Therban® 3627 acrylonitrile units: 36% by weight — Materials Used - Therban® 3627 Mooney viscosity: ML 1+4@100° C. 66 — Materials Used - Therban® 3627 gel content: 0.7% — Materials Used - Therban® 3627 noble metal content: 2 ppm rhodium, <1 ppm ruthenium, <1 ppm palladium — Materials Used - Therban® 3627 acetone concentration: 10% — Performance of Ageing Investigations - Mixing time: 2 hours — Performance of Ageing Investigations - Drying temperature: 55° C. — Performance of Ageing Investigations - Ageing temperature: 140° C. — Mooney Ageing / Performance of Ageing Investigations - Ageing time: 4 days — Mooney Ageing / Performance of Ageing Investigations - MEK volume: approximately 20 ml — Gel Content - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - degree of hydrogenation: Not disclosed in extracted evidence

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Solution polymerization / solution hydrogenation

degree of hydrogenation - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1-17: HNBR compositions containing different noble-metal compounds were prepared, dried, and aged to assess gel content and Mooney viscosity changes. - Table 1: Ruthenium compounds showed relatively small ageing effects compared with palladium compounds, based on gel content and Mooney viscosity changes. - Table 1: The reference HNBR had 0.7% gel content and Mooney viscosity 66 before ageing. - Table 1: After 4 days at 140° C., the reference HNBR increased to 4.7% gel content and 81 Mooney viscosity. - Table 1: The patent links catalyst residue type and level to ageing performance of HNBR compositions.

Technical Relevance This patent is relevant because it addresses HNBR produced by solution hydrogenation and focuses on how catalyst choice and residue levels affect ageing, a key quality issue in HNBR manufacture. It provides explicit catalyst loading and operating-condition ranges for producing HNBR with improved stability.

Patent 3

- Patent Number: EP3827029A1
- Patent Title: Method for producing hydrogenated nitrile rubber and hnbr compositions thereof
- Assignee: ARLANXEO Deutschland GmbH
- Jurisdiction: EP
- Publication Year: 2021
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses solution hydrogenation of at least partially unsaturated nitrile rubber in the presence of ruthenium, palladium, or rhodium compounds, with emphasis on producing HNBR having good ageing properties and on catalyst residue management. The extracted evidence is claim- and description-based and does not include a worked example section. - Ruthenium amount: 10 ppm to 200 ppm — Claim 1 - Ruthenium amount: 10 ppm to 150 ppm — Claim 1 - Ruthenium amount: 10 ppm to 120 ppm — Claim 1 - Ruthenium amount: 10 ppm to 79 ppm — Claim 1 - Ruthenium amount: 42 ppm to 79 ppm — Claim 1 - Palladium amount: 20 ppm to <67 ppm — Claim 1 - Palladium amount: 44 ppm to <67 ppm — Claim 1 - Rhodium amount: 50 ppm to <270 ppm — Claim 1 - Rhodium amount: 79 ppm to <270 ppm — Claim 1 - Rhodium amount: 50 ppm to 250 ppm — Claim 1 - Rhodium amount: 50 ppm to 170 ppm — Claim 1 - Rhodium amount: 79 ppm to 170 ppm — Claim 1 - Hydrogenation temperature: 60 to 200° C. — Claim 1 - Hydrogenation pressure: 700 000 pascals to 15 000 000 pascals — Claim 1 - Hydrogenation time: 1 to 24 hours — Claim 1 - Therban® 3627 acrylonitrile units: 36% by weight — Materials Used - Therban® 3627 Mooney viscosity: ML 1+4@100° C. 66 — Materials Used - Therban® 3627 gel content: 0.7% — Materials Used - Therban® 3627 noble metal content: 2 ppm rhodium, <1 ppm ruthenium, <1 ppm palladium — Materials Used - Therban® 3627 acetone concentration: 10% — Performance of Ageing Investigations - Mixing time: 2 hours — Performance of Ageing Investigations - Drying temperature: 55° C. — Performance of Ageing Investigations - Ageing temperature: 140° C. — Mooney Ageing / Performance of Ageing Investigations - Ageing time: 4 days — Mooney Ageing / Performance of Ageing Investigations - MEK volume: approximately 20 ml — Gel Content - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - degree of hydrogenation: Not disclosed in extracted evidence

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Solution polymerization / solution hydrogenation

degree of hydrogenation - Not disclosed in extracted evidence

Relevant Experimental Evidence - No worked examples were detected by the parser; the evidence is drawn from claims and detailed description. - The description states that HNBR is produced by hydrogenation of NBR in solution with Ru, Pd, or Rh compounds. - The patent emphasizes improved ageing properties and reduced need for catalyst-removal steps. - The description cites prior-art catalyst-removal methods and positions the invention as avoiding those complications. - The extracted evidence includes explicit catalyst loading and operating-condition ranges for the hydrogenation step.

Technical Relevance This patent is directly relevant because it defines a solution hydrogenation route to HNBR and ties catalyst residue control to ageing performance. It is useful for understanding how HNBR can be produced with noble-metal catalysts while maintaining product stability.

Patent 4

- Patent Number: US9150669B2

- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Lanxess Deutschland GmbH
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses simultaneous hydrogenation and metathesis of nitrile rubber in monochlorobenzene solution using ruthenium/osmium-based catalysts, with the goal of producing low-molecular-weight HNBR having narrow molecular-weight distribution. The examples compare Hoveyda, Arlt, and Wilkinson-type catalysts and show that the inventive catalysts accelerate hydrogenation while reducing molecular weight. - NBR concentration: 5.7 wt % — Comparative Examples / Inventive Examples - NBR solution mass: 518 g — Examples 1-2 - Monochlorobenzene mass: 4300 g — Examples 1-2 - Room temperature: 22+/-2° C. — Abbreviations - Agitation speed: 600 rpm — Examples 1-2 - Degassing pressure: 100 psi H₂ — Examples 1-2 - Degassing pressure: 85 bar — Examples 1-2 - Reaction temperature: 130° C. — Examples 1-2 - Reaction temperature: 138° C. — Examples 1-2 - Reaction pressure: 85 bar — Examples 1-2 - Reaction time: 240 min — Table 2 / Example 1 - Catalyst amount: 0.041 phr — Table 1 - Catalyst amount: 0.045 phr — Table 1 - Catalyst amount: 0.060 phr — Table 1 - Metal content: 65.6 ppm — Table 1 - Co-catalyst amount: 1.0 phr — Table 1 - Reaction time: 60 min — Table 2 - Reaction time: 120 min — Table 2 - Reaction time: 180 min — Table 2 - Reaction time: 240 min — Table 2 - RDB: 2.5% — Table 2 - RDB: 7.0% — Table 2 - RDB: 15.4% — Table 2 - RDB: 0.6% — Table 2 - RDB: 4.5% — Table 2 - RDB: 3.1% — Table 2 - RDB: 0.5% — Table 2 - RDB: 3.9% — Table 2 - RDB: 1.3% — Table 2 - RDB: 0.4% — Table 2 - RDB: 3.5% — Table 2 - RDB: 0.7% — Table 2 - Mw: 113371 g/mol — Table 3 - Mn: 48352 g/mol — Table 3 - PDI: 2.34 — Table 3 - Mooney viscosity: 33.3 — Table 3 - Mw: 133758 g/mol — Table 3 - Mn: 54595 g/mol — Table 3 - PDI: 2.45 — Table 3 - Mooney viscosity: 41.3 — Table 3 - Mw: 195304 g/mol — Table 3 - Mn: 64130 g/mol — Table 3 - PDI: 3.05 — Table 3 - Mooney viscosity: 69.3 — Table 3 - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - degree of hydrogenation: RDB 0.4% to 0.7% after 240 min in the inventive examples; 0.7% in the comparison example

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Organic solvent / solution polymerization in monochlorobenzene

degree of hydrogenation - RDB 0.4% to 0.7% after 240 min in the inventive examples; 0.7% in the comparison example

Relevant Experimental Evidence - Example 1: Hoveyda and Arlt catalysts gave faster hydrogenation and lower final molecular weight than Wilkinson's catalyst. - Table 2: Residual double bond content dropped to 0.4-0.7% after 240 min for the inventive catalysts, versus 0.7% for Wilkinson's catalyst. - Table 3: The inventive catalysts produced lower Mw and lower Mooney viscosity than the comparison catalyst. - The patent demonstrates simultaneous

hydrogenation/metathesis as a route to low-molecular-weight HNBR. - The catalyst system is designed to avoid the need for a separate metathesis step and to control molecular-weight distribution.

Technical Relevance This is a key HNBR manufacturing patent because it combines hydrogenation with metathesis to tune both saturation and molecular weight. It is highly relevant to HNBR synthesis strategy, especially where low viscosity and narrow molecular-weight distribution are desired.

Patent 5

- Patent Number: US10000584B2
- Patent Title: Hydrogenation of nitrile rubber
- Assignee: Arlanxeo Deutschland Gmbh
- Jurisdiction: US
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - degree of hydrogenation: Not disclosed in extracted evidence

Medium & Water Role - No water-related process step disclosed in the extracted evidence.

degree of hydrogenation - Not disclosed in extracted evidence

Relevant Experimental Evidence - Comparative Examples: Comparative Examples - Comparative Example: Comparative Example - EXAMPLES: EXAMPLES - Inventive Examples: Inventive Examples

Technical Relevance Selected via deterministic pipeline scoring.

Patent 6

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: LANXESS Deutschland GmbH
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses simultaneous hydrogenation and metathesis of nitrile rubber in monochlorobenzene solution using ruthenium/osmium-based catalysts, producing low-molecular-weight HNBR with narrow molecular-weight distribution. The examples compare Hoveyda, Arlt, and Wilkinson-type catalysts and show that the inventive catalysts improve hydrogenation efficiency while lowering molecular weight. - NBR concentration: 5.7 wt % — Examples 1-2 - NBR solution mass: 518 g — Examples 1-2 - Monochlorobenzene mass: 4300 g — Examples 1-2 - Room temperature: 22+/-2° C. — Abbreviations - Agitation speed: 600 rpm — Examples 1-2 - Degassing pressure: 100 psi H2 — Examples 1-2 - Reaction temperature: 130° C. — Examples 1-2 - Reaction temperature: 138° C. — Examples 1-2 - Reaction pressure: 85 bar — Examples 1-2 - Catalyst amount: 0.041 phr — Table 1 - Catalyst amount: 0.045 phr — Table 1 - Catalyst amount: 0.060 phr — Table 1 - Metal content: 65.6 ppm — Table 1 - Co-catalyst amount: 1.0 phr — Table 1 - Reaction time: 60 min — Table 2 - Reaction time: 120 min — Table 2 - Reaction time: 180 min — Table 2 - Reaction time: 240 min — Table 2 - RDB: 2.5% — Table 2 - RDB: 7.0% — Table 2 - RDB: 15.4% — Table 2 - RDB: 0.6% — Table 2 - RDB: 4.5% — Table 2 - RDB: 3.1% — Table 2 - RDB: 0.5% — Table 2 - RDB: 3.9% — Table 2 - RDB: 1.3% — Table 2 - RDB: 0.4% — Table 2 - RDB: 3.5% — Table 2 - RDB: 0.7% — Table 2 - Mw: 113371 g/mol — Table 3 - Mn: 48352 g/mol — Table 3 - PDI: 2.34 — Table 3 - Mooney viscosity: 33.3 — Table 3 - Mw: 133758 g/mol — Table 3 - Mn: 54595 g/mol — Table 3 - PDI: 2.45 — Table 3 - Mooney viscosity: 41.3 — Table 3 - Mw: 195304 g/mol — Table 3 - Mn: 64130 g/mol — Table 3 - PDI: 3.05 — Table 3 - Mooney viscosity: 69.3 — Table 3 - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - degree of hydrogenation: RDB 0.4% to 0.7% after 240 min in the inventive examples; 0.7% in the comparison example

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Organic solvent / solution polymerization in monochlorobenzene

degree of hydrogenation - RDB 0.4% to 0.7% after 240 min in the inventive examples; 0.7% in the comparison example

Relevant Experimental Evidence - Example 1: Hoveyda and Arlt catalysts outperformed Wilkinson's catalyst in hydrogenation rate and molecular-weight reduction. - Table 2: Residual double bond content fell to 0.4-0.7% after 240 min for the inventive catalysts. - Table 3: The inventive catalysts yielded lower Mw and lower Mooney viscosity than the comparison catalyst. - The patent demonstrates a one-pot hydrogenation/metathesis route to low-Mw HNBR. - The process is explicitly framed as a way to avoid separate metathesis and hydrogenation steps.

Technical Relevance This patent is directly relevant to HNBR synthesis because it produces hydrogenated nitrile rubber while simultaneously controlling molecular weight through metathesis. It provides a practical route to low-viscosity HNBR with very low residual unsaturation.

Patent 7

- Patent Number: US7846995B2
- Patent Title: Process for removing ruthenium-containing catalyst residues from optionally ...

- Assignee: Lanxess Deutschland GmbH
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses post-treatment of optionally hydrogenated nitrile rubber solutions with functionalized ion-exchange resins to remove ruthenium-containing catalyst residues. The HNBR itself is not synthesized in this patent; instead, the focus is on purification of an already hydrogenated nitrile rubber solution and recovery of low-ruthenium product. - Average particle diameter: 0.2 to 2.5 mm dry basis — Claim 1 - Ruthenium content: 5 to 1000 ppm — Claim 2 - Ruthenium content: 5 to 500 ppm — Claim 3 - HNBR concentration: 0.5 to 30% by weight — Claim 4 - Reaction temperature: about 100° C. — Example 1A - Reaction time: 66 hours — Example 1A - Resin mass: 0.5 g — Example 1A - Rubber solution mass: 180 g — Example 1A - Ru content after treatment: 10 ppm — Example 1A - Initial Ru content: 15 ppm — Example 1A - Removal: 33% — Example 1A - Resin mass: 0.5 g — Example 2A - Rubber solution mass: 180 g — Example 2A - Reaction temperature: about 100° C. — Example 2A - Reaction time: 66 hours — Example 2A - Ru content after treatment: 10 ppm — Example 2A - Initial Ru content: 15 ppm — Example 2A - Removal: 33% — Example 2A - Hydrogenated nitrile rubber acrylonitrile content: 34% b.w. — Examples - Residual double bonds: less than 0.9% — Examples - Mooney viscosity: about 40 — Examples - Standard rubber solution concentration: 6.0% by weight — Examples - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - degree of hydrogenation: Not disclosed in extracted evidence

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Organic solvent / solution treatment in monochlorobenzene

degree of hydrogenation - Not disclosed in extracted evidence

Relevant Experimental Evidence - Example 1A: Thiourea-functionalized resin reduced Ru content from 15 ppm to 10 ppm in the HNBR solution. - Example 2A: Amine-functionalized resin gave the same 33% Ru removal under similar conditions. - The patent demonstrates purification of hydrogenated nitrile rubber rather than hydrogenation itself. - The extracted evidence shows solvent recovery by evaporation and drying after resin treatment. - The starting HNBR is described as having less than 0.9% residual double bonds and Mooney viscosity about 40.

Technical Relevance This patent is relevant to HNBR manufacturing because it addresses removal of ruthenium residues from hydrogenated nitrile rubber solutions, a critical downstream step for product quality and ageing performance. It does not disclose the hydrogenation synthesis itself, but it is important for finishing HNBR made by hydrogenation routes.

Patent 8

- Patent Number: EP2868676A1
- Patent Title: Functionalised copolymer rubber containing nitrile groups
- Assignee: LANXESS Deutschland GmbH
- Jurisdiction: EP
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses functionalized nitrile-group-containing copolymer rubbers, including hydrogenated variants, with specified monomer compositions and hydrogenation levels. The examples focus on characterization and vulcanizate performance of HNBR-based materials rather than a detailed hydrogenation recipe. - Nitrile unit content: 10% to 60% by weight — Claim 1 - Nitrile unit content: 20% to 50% by weight — Claim 1 - Nitrile unit content: 34% to 43% by weight — Claim 1 - Conjugated diene unit content: 31% to 89.0% by weight — Claim 1 - Conjugated diene unit content: 42% to 78.75% by weight — Claim 1 - Conjugated diene unit content: 49% to 64.5% by weight — Claim 1 - Carbonyl-containing monomer unit content: 1% to 9% by weight — Claim 1 - Carbonyl-containing monomer unit content: 1.25% to 8% by weight — Claim 1 - Carbonyl-containing monomer unit content: 1.5% to 8% by weight — Claim 1 - Hydrogenation level: 50% or more — Claim 1 - Hydrogenation level: 75% or more — Claim 1 - Hydrogenation level: 85% or more — Claim 1 - HNBR 1 ACN content: 33.6% — Inventive examples - HNBR 1 MEA content: 4.8% — Inventive examples - HNBR 1 Mooney viscosity: 67 — Inventive examples - HNBR 2 ACN content: 34.6% — Inventive examples - HNBR 2 PEG5 content: 4.1% — Inventive examples - HNBR 2 Mooney viscosity: 88 — Inventive examples - HNBR 3 ACN content: 33.6% — Inventive examples - HNBR 3 MEA content: 7.7% — Inventive examples - HNBR 3 Mooney viscosity: 72 — Inventive examples - HNBR 4 ACN content: 34% — Inventive examples - HNBR 4 residual double bond content: less than 0.9% — Inventive examples - HNBR 4 Mooney viscosity: 70 — Inventive examples - Mixing roll mill roll diameter: 80 mm — Example - Mixing roll mill roll length: 200 mm — Example - Roll preheat temperature: 40°C — Example - Front roll speed: 16.5 rpm — Example - Rear roll speed: 20 rpm — Example - Friction ratio: 1:1.2 — Example - Mixing time: 5 to 8 minutes — Example - Vulcanization temperature: 180°C — Example - MDR angle: 0.5° — Example - Oscillation frequency: 1.7 Hz — Example - MDR time: 30 minutes — Example - Plaque thickness: 2 mm — Example - Hot-air ageing: 168 h at 150°C — Example - IRM 903 ageing: 168 h at 150°C — Example - DSC heating rate: 20 K/minute — Example - DSC heating range: -100°C to 100°C — Example - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - degree of hydrogenation: 50% or more, preferably 75% or more and more preferably 85% or more

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Not disclosed for synthesis in the extracted evidence; formulation/compounding in a mixing roll mill is disclosed

degree of hydrogenation - 50% or more, preferably 75% or more and more preferably 85% or more

Relevant Experimental Evidence - Inventive examples define HNBR 1-3 with different termonomer contents and Mooney viscosities. - HNBR 4 is a commercial hydrogenated nitrile-butadiene copolymer with less than 0.9% residual double bonds. - The examples focus on compounding and vulcanizate performance rather than synthesis of the HNBR itself. - The patent shows that hydrogenated nitrile copolymers can be functionalized with methoxyethyl acrylate or PEG acrylate units. - The disclosed hydrogenation level threshold is at least 50%, preferably 75% or 85%.

Technical Relevance This patent is relevant because it explicitly includes hydrogenated nitrile copolymer rubbers and defines hydrogenation level thresholds for the target material class. It is more formulation-focused than synthesis-focused, but it provides useful target-property context for HNBR materials.

Patent 9

- Patent Number: EP3063192B1
- Patent Title: Copolymer rubber containing nitrile groups
- Assignee: ARLANXEO Deutschland GmbH
- Jurisdiction: EP
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses hydrogenated nitrile-butadiene copolymers containing additional acrylate-type termonomers, with the hydrogenation level defined as part of the claimed material. The examples are compounding and performance tests on already hydrogenated polymers rather than a detailed hydrogenation synthesis procedure. - Nitrile unit content: 10% to 60% by weight — Claim 1 - Nitrile unit content: 20% to 50% by weight — Claim 1 - Nitrile unit content: 34% to 43% by weight — Claim 1 - Conjugated diene unit content: 31% to 89.0% by weight — Claim 1 - Conjugated diene unit content: 42% to 78.75% by weight — Claim 1 - Conjugated diene unit content: 49% to 64.5% by weight — Claim 1 - Termonomer content: 1% to 9% by weight — Claim 1 - Termonomer content: 1.25% to 8% by weight — Claim 1 - Termonomer content: 1.5% to 8% by weight — Claim 1 - Hydrogenation level: 50% or more — Claim 1 - Hydrogenation level: 75% or more — Claim 1 - Hydrogenation level: 85% or more — Claim 1 - HNBR 1 ACN content: 33.6% — Inventive examples - HNBR 1 MEA content: 4.8% — Inventive examples - HNBR 1 Mooney viscosity: 67 — Inventive examples - HNBR 2 ACN content: 34.6% — Inventive examples - HNBR 2 PEG5 content: 4.1% — Inventive examples - HNBR 2 Mooney viscosity: 88 — Inventive examples - HNBR 3 ACN content: 33.6% — Inventive examples - HNBR 3 MEA content: 7.7% — Inventive examples - HNBR 3 Mooney viscosity: 72 — Inventive examples - HNBR 4 ACN content: 34% —

Inventive examples - HNBR 4 residual double bond content: less than 0.9% — Inventive examples - HNBR 4 Mooney viscosity: 70 — Inventive examples - Mixing roll mill roll diameter: 80 mm — Example - Mixing roll mill roll length: 200 mm — Example - Roll preheat temperature: 40°C — Example - Front roll speed: 16.5 rpm — Example - Rear roll speed: 20 rpm — Example - Friction ratio: 1:1.2 — Example - Mixing time: 5 to 8 minutes — Example - Vulcanization temperature: 180°C — Example - MDR angle: 0.5° — Example - Oscillation frequency: 1.7 Hz — Example - MDR time: 30 minutes — Example - Plaque thickness: 2 mm — Example - Hot-air ageing: 168 h at 150°C — Example - IRM 903 ageing: 168 h at 150°C — Example - DSC heating rate: 20 K/minute — Example - DSC heating range: -100°C to 100°C — Example - Medium & Water Role: No water-related process step disclosed in the extracted evidence. - degree of hydrogenation: 50% or more, preferably 75% or more and more preferably 85% or more

Medium & Water Role - No water-related process step disclosed in the extracted evidence. - Core reaction medium: Not disclosed for synthesis in the extracted evidence; formulation/compounding in a mixing roll mill is disclosed

degree of hydrogenation - 50% or more, preferably 75% or more and more preferably 85% or more

Relevant Experimental Evidence - Inventive examples define HNBR 1-3 with different termonomer contents and Mooney viscosities. - HNBR 4 is a commercial hydrogenated nitrile-butadiene copolymer with less than 0.9% residual double bonds. - The examples focus on compounding and performance testing rather than synthesis of the HNBR itself. - The patent shows that hydrogenated nitrile copolymers can be functionalized with methoxyethyl acrylate or PEG acrylate units. - The disclosed hydrogenation level threshold is at least 50%, preferably 75% or 85%.

Technical Relevance This patent is relevant because it defines hydrogenated nitrile-butadiene copolymers with additional functional monomers and explicitly states hydrogenation-level thresholds. It supports the broader HNBR target space, although the extracted evidence is primarily formulation-oriented.

PATENT COMPARISON TABLE

Patent	Applicant	Priority	Medium & Water Role	degree of hydrogenation	Key Findings

US9657113B2	University Of Waterloo	2017	The core hydrogenation medium is an aqueous NBR latex suspension; water is part of the reaction medium, and in some examples additional water is added for dilution. No separate downstream water workup is emphasized in the extracted evidence.	98%	Example 1: Direct latex hydrogenation reached 98% hydrogenation in 4 hours with no visible gel and produced soluble HNBR
US20210340285A1	Arlanxeo Deutschland GmbH	2021	No water-related process step disclosed in the extracted evidence.	Not disclosed in extracted evidence	Example 1-17: HNBR compositions containing different noble-metal compounds were prepared, dried, and aged to assess gel

EP3827029A1	ARLANXEO Deutschland GmbH	2021	No water-related process step disclosed in the extracted evidence.	Not disclosed in extracted evidence	No worked examples were detected by the parser; the evidence is drawn from claims and detailed description.
US9150669B2	Lanxess Deutschland GmbH	2015	No water-related process step disclosed in the extracted evidence.	RDB 0.4% to 0.7% after 240 min in the inventive examples; 0.7% in the comparison example	Example 1: Hoveyda and Arlt catalysts gave faster hydrogenation and lower final molecular weight than Wilkinson's cataly
US10000584B2	Arlanxio Deutschland GmbH	2018	No water-related process step disclosed in the extracted evidence.	Not disclosed in extracted evidence	Comparative Examples: Comparative Examples

EP2473281B1	LANXESS Deutschland GmbH	2014	No water-related process step disclosed in the extracted evidence.	RDB 0.4% to 0.7% after 240 min in the inventive examples; 0.7% in the comparison example	Example 1: Hoveyda and Arlt catalysts outperformed Wilkinson's catalyst in hydrogenation rate and molecular-weight reduc
US7846995B2	Lanxess Deutschland GmbH	2010	No water-related process step disclosed in the extracted evidence.	Not disclosed in extracted evidence	Example 1A: Thiourea-functionalized resin reduced Ru content from 15 ppm to 10 ppm in the HNBR solution.
EP2868676A1	LANXESS Deutschland GmbH	2015	No water-related process step disclosed in the extracted evidence.	50% or more, preferably 75% or more and more preferably 85% or more	Inventive examples define HNBR 1-3 with different termonomer contents and Mooney viscosities.

EP3063192B1	ARLANXEO Deutschland GmbH	2018	No water-related process step disclosed in the extracted evidence.	50% or more, preferably 75% or more and more preferably 85% or more	Inventive examples define HNBR 1-3 with different termonomer contents and Mooney viscosities.
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3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US9657113B2 is the only aqueous-latex hydrogenation patent in the set; the others with direct synthesis evidence are solution-phase in monochlorobenzene or acetone-based solution handling.
- US9657113B2 achieves 98-99% hydrogenation in 0.5-10 h depending on catalyst loading, temperature, and pressure, while US9150669B2 / EP2473281B1 report residual double-bond contents down to 0.4-0.7% after 240 min in simultaneous hydrogenation/metathesis.
- US20210340285A1 and EP3827029A1 emphasize catalyst residue control and ageing performance rather than reporting a specific hydrogenation degree for the target HNBR in the extracted evidence.
- US7846995B2 is a downstream purification patent: it treats already hydrogenated nitrile rubber solutions to remove Ru residues, not to perform hydrogenation.
- EP2868676A1 and EP3063192B1 are formulation/property patents that define hydrogenation thresholds for HNBR-type copolymers (50% or more, preferably 75% or 85% or more) but do not disclose a synthesis route in the extracted evidence.
- The metathesis-assisted patents (US9150669B2 and EP2473281B1) uniquely couple hydrogenation with molecular-weight reduction, yielding lower Mw and lower Mooney viscosity than the comparison catalyst system.

4. CONCLUSION

Across the selected primary patents, HNBR is produced by two dominant routes: direct hydrogenation of NBR latex in water and solution-phase hydrogenation in organic solvent, including one-pot hydrogenation/metathesis systems that simultaneously lower molecular weight. The strongest synthesis disclosures are US9657113B2, which demonstrates near-quantitative

latex hydrogenation with HG2 catalyst, and US9150669B2 / EP2473281B1, which show that Ru/Os catalyst systems can drive residual double bonds to very low levels while reducing Mw and Mooney viscosity. Other patents in the set focus on catalyst-residue control, ageing stability, or formulation/property definitions for hydrogenated nitrile copolymers, reinforcing that HNBR manufacturing is tightly coupled to downstream purification and performance requirements. The extracted evidence supports a broad target hydrogenation window from at least 50% to near-complete saturation, with practical examples reaching 98-99% hydrogenation or residual double-bond contents below 1%.

5. REFERENCES

- US9657113B2 | Hydrogenation of a diene-based polymer latex | University Of Waterloo | US | 2017 | <https://patents.google.com/patent/US9657113B2/en>
- US20210340285A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | Arlanxeo Deutschland GmbH | US | 2021 | <https://patents.google.com/patent/US20210340285A1/en>
- EP3827029A1 | Method for producing hydrogenated nitrile rubber and hnbr compositions thereof | ARLANXEO Deutschland GmbH | EP | 2021 | <https://patents.google.com/patent/EP3827029A1/en>
- US9150669B2 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | Lanxess Deutschland GmbH | US | 2015 | <https://patents.google.com/patent/US9150669B2/en>
- US10000584B2 | Hydrogenation of nitrile rubber | Arlanxeo Deutschland GmbH | US | 2018 | <https://patents.google.com/patent/US10000584B2/en>
- EP2473281B1 | Process for the preparation of low molecular weight hydrogenated nitrile rubber | LANXESS Deutschland GmbH | EP | 2014 | <https://patents.google.com/patent/EP2473281B1/en>
- US7846995B2 | Process for removing ruthenium-containing catalyst residues from optionally ... | Lanxess Deutschland GmbH | US | 2010 | <https://patents.google.com/patent/US7846995B2/en>
- EP2868676A1 | Functionalised copolymer rubber containing nitrile groups | LANXESS Deutschland GmbH | EP | 2015 | <https://patents.google.com/patent/EP2868676A1/en>
- EP3063192B1 | Copolymer rubber containing nitrile groups | ARLANXEO Deutschland GmbH | EP | 2018 | <https://patents.google.com/patent/EP3063192B1/en>